

Advanced Energy's SystemVision

Advanced Energy is a Raleigh, North Carolina-based nonprofit serving as a North Carolina and national resource that focuses on industrial process technologies, motors and drives testing, and applied building science. Its facility houses state-of-the-art laboratories for testing and applied research in all three of these evolving disciplines. Advanced Energy creates economic, environmental, and societal benefits through innovative and market-based approaches to energy issues.

Launched in 2001, SystemVision, Advanced Energy's affordable housing program, provides the training and technical support that leads to the improved health, safety, durability, comfort, and energy efficiency of affordable homes. Nonprofits developing low- and moderate-income housing within North Carolina are eligible to start building with SystemVision, and, through the North Carolina Housing Finance Agency, funding is available to these nonprofit organizations to help defray the costs often associated with building high-performance homes. Self Help Durham is one such affordable housing program. SystemVision trains builders, and insulation and HVAC contractors who build homes for Self Help; and they inspect the home after framing, after insulation, and at completion, making sure that heating and cooling ducts are tight.

"We sell homes to families that are former renters paying up to \$300 a month for heating their homes," says Roger Chiles, director of Self Help Durham. "With our SystemVision homes, our clients may only pay \$35 a month for heating, in a three-bedroom, two-bath home." And the homes continue to perform well and help low-income families meet day-to-day living expenses—heating and cooling costs can be the second most expensive monthly bill, after their mortgages. "Advanced

Energy guarantees low utility bills for 2 years, but the way we build houses under their guidance keeps them performing well for 10 or 15 years or longer," says Chiles. "And it takes just a phone call to get someone out to the building site if we have a problem. I think these guys are great."

To date, Advanced Energy has built 1,200 homes in its SystemVision program. And homes built in the program have been proven to save 15% more in energy use and cost than standard code-built homes. After performing an energy use study comparing Energy Star Homes, Environments for Living homes, and code-built homes in Phoenix, Arizona, Advanced Energy decided to quantify the performance of its low-income program. A study was formed comparing 156 SystemVision homes with 440 code-built homes in central North Carolina. The study assessed 12 months of energy usage data in homes that were less than five years old. While a typical 1,700 ft² house used approximately 17,056 kWh of energy at a total annual cost of \$1,431, the homes built to SystemVision standards used approximately 14,579 kWh of energy at a total annual cost of \$1,223, a savings of more than \$200, or nearly 15%. This documentation is significant, for home energy savings are often based on computer modeling rather than on verified data.

Perhaps more important than the lower average energy use per home, the SystemVision homes sampled showed a smaller range of energy use and cost than the sample of code-built homes. Eliminating the energy hogs in the marketplace, as this study showed, is a crucial step toward making the operation, not just the purchase price, of homes affordable. Furthermore, while this study showed 30% savings in heating and cooling energy use and cost, resulting in the 15% savings overall, the overall energy use and cost of SystemVision home operation is expected to drop, as SystemVision homes built since the summer of 2006 are held to higher energy efficiency standards than those studied. These new standards require an Energy Star-

labeled major appliance and CFL bulbs in the home, among other measures.

"Our goal is to improve energy efficiency in every home receiving public or charitable funding, at least to Energy Star level, while improving the homes' safety, durability, comfort, affordability, and environmental impact," says Brian Coble, director of high-performance homes at Advanced Energy.

Advanced Energy provides ongoing support throughout the design, construction, and marketing processes, and all homes in North Carolina's SystemVision program carry a heating-and-cooling energy guarantee. These house plan-specific guarantees generally range from \$16 to \$40 per month, with most falling between \$25 and \$30. If a home exceeds its guaranteed usage at the end of the year, Advanced Energy pays the difference to the homeowner. While the program has paid out to homeowners, Advanced Energy believes that this shows the need for the guarantee. In seven years the program has paid out just over \$3,500, with checks as low as \$0.50 and as high as \$250. "The majority of our high payouts tend to come from equipment failure, such as leaking refrigerant or improper use of the HVAC systems thermostat," says Krista Egger, SystemVision's program manager.

In addition to paying out, if the home exceeds its guarantee amount by \$75, Advanced Energy visits the home to find the problem. SystemVision homes also carry an Advanced Energy comfort guarantee to ensure that the center of each room stays within 3°F of the thermostat setting. The comfort guarantee makes certain that the homes are not only financially affordable to purchase and live in, but comfortable and enjoyable too.

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—Brian Coble and Krista Egger

Brian Coble is the director of affordable housing and Krista Egger is an affordable housing program manager at Advanced Energy.

For more information:

To learn more about Advanced Energy and the SystemVision program, visit www.advancedenergy.org.